

Tanmay Ambadkar

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EDUCATION

The Pennsylvania State University
Ph.D. in Computer Science and Engineering

University Park, PA
Jan. 2024 – May 2027 (Expected)

EXPERIENCE

The Pennsylvania State University
Research Assistant

University Park, PA
May 2023 – Present

- Developing and refining specification-guided frameworks to enhance the reliability, safety, and interpretability of RL agents.
- Designing frameworks to implement Reinforcement Learning agents for optimizing operational costs of grid-integrated district energy systems, achieving a **32% cost reduction**.
- Predicted Autism Spectrum Disorder cases in children using time-series models on Electronic Health Records (EHR) data, preprocessing over 500GB of data with PySpark.

Siemens Technology and Services

Remote

Research and Digitization Automation Intern

Jan. 2022 – July 2022

- Owned the end-to-end development of a plug-and-play anomaly detection library using AutoEncoders and Explainable AI, deploying the final model on Databricks and presenting a working demo to project stakeholders.
- Engineered end-to-end forecasting workflows that achieved a **0.98 r^2 score** on internal benchmarks and built CNN-based models for a major client (Starbucks).

Siemens Technology and Services

Remote

Research and Digitization Automation Intern

May 2021 – July 2021

- Optimized the Industrial Predictive Analytics Engine (IPAE) by integrating real-time monitoring tools, reducing pipeline execution time by **30%**.
- Developed a library to forecast building occupancy and energy requirements in conjunction with an RL model for optimizing building setpoints for maximum energy savings, for Dubai Expo 2021.

PUBLICATIONS

Robust Adaptive Multi-Step Predictive Shielding. T. Ambadkar, D. Chudiwal, G. Anderson, and A. Verma. *Accepted at ICLR 2026; Student Abstract at AAAI 2026.* [\[Paper\]](#) [\[Project Page\]](#)

AutoSpec: Automating the Refinement of Reinforcement Learning Specifications. T. Ambadkar, Djordje Zikelic, and A. Verma. *Accepted at ICLR 2026; P-AI-FM Workshop at AAAI-2026; Poster at PLDI Student Research Competition, 2024.* [\[Paper\]](#)

Preference Conditioned Multi-Objective Reinforcement Learning. T. Ambadkar, S. Panda, S. Kale, A. Verma, and J. Dodge. *Submitted to ICLR 2026.* [\[Paper\]](#) [\[Project Page\]](#)

Specification Guided Reinforcement Learning. T. Ambadkar, *Accepted at AAAI Doctoral Consortium 2026*

Safer Policies via Affine Representations using Koopman Dynamics. T. Ambadkar, D. Chudiwal, G. Anderson, and A. Verma. *Submitted to AAAI 2026.*

MIXTAPE: Middleware for Interactive XAI with Tree-Based AI Performance Evaluation. T. Ambadkar, H. Moore, S. Panda, et al. *SISO SIMposium, 2025.*

Optimizing Operational Costs in Combined Heat and Power Integrated District Heating Systems. S. Anbarasu, T. Ambadkar, R. Adhikari, et al. *SimBuild, 2024.* [\[Paper\]](#)

Deep reinforcement learning approach to predict head movement in 360° videos. T. Ambadkar and P. Mazumdar. *Proc. IS&T Int'l. Symp. on Electronic Imaging, 2022.* [\[Paper\]](#) [\[Code\]](#)

TECHNICAL SKILLS

Programming Languages: Python, PySpark, JavaScript, C++

Frameworks: PyTorch, TensorFlow, Stable-Baselines3, Scikit-learn, Flask

Topics: Reinforcement Learning, AI Safety, Formal Methods, Multi-Objective RL, Time-Series Analysis